

## PORT ARANSAS PIER, TX, USA

WMO: 994110

Lat: 27.8259N

Lon: 97.0506W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 37.4       | 41.3 | 23.8                          | 18.0 | 48.2 | 28.4 | 22.3 | 49.7 | 40.5                  | 52.4 | 34.5 | 57.2 | 18.8                  | 0    | 0.623 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.0                    | 86.2            | 78.2 | 85.5 | 78.6 | 85.1 | 78.5 | 81.5                | 84.2 | 80.7 | 83.9 | 80.2 | 83.7 | 13.3                 | 140  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.6                            | 158.7 | 83.7 | 79.8 | 154.6 | 83.4 | 79.1 | 150.8 | 83.2 | 45.1          | 84.3 | 44.3 | 84.0 | 43.6 | 83.8 | 84.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 39.0              | 32.7 | 27.0 | 33.8                       | 88.8 | 2.4                | 2.6 | 32.0                                               | 90.6 | 30.6       | 92.1 | 29.2       | 93.6 | 27.5       | 95.4 |
|                   |      |      | 31.2                       | 82.0 | 1.9                | 1.5 | 29.8                                               | 83.0 | 28.7       | 83.9 | 27.6       | 84.7 | 26.3       | 85.7 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 71.4   | 57.0  | 58.8  | 63.5  | 70.5  | 76.5  | 81.2  | 82.3  | 82.9  | 81.0  | 75.5  | 66.8  | 59.7 |
|                                                                               | DBStd       | 10.77  | 7.40  | 6.66  | 5.85  | 4.31  | 2.97  | 1.78  | 1.48  | 1.52  | 3.23  | 5.61  | 7.48  | 7.74 |
|                                                                               | HDD50       | 68     | 29    | 15    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 16   |
|                                                                               | HDD65       | 805    | 253   | 180   | 90    | 11    | 0     | 0     | 0     | 0     | 0     | 7     | 69    | 195  |
|                                                                               | CDD50       | 7877   | 247   | 262   | 423   | 615   | 823   | 936   | 1001  | 1021  | 931   | 791   | 509   | 318  |
|                                                                               | CDD65       | 3136   | 5     | 7     | 45    | 176   | 358   | 486   | 536   | 556   | 481   | 332   | 123   | 31   |
|                                                                               | CDH74       | 29871  | 1     | 4     | 9     | 274   | 2401  | 5437  | 6352  | 6877  | 5539  | 2694  | 279   | 4    |
|                                                                               | CDH80       | 7876   | 0     | 1     | 1     | 16    | 71    | 1293  | 1955  | 2459  | 1737  | 341   | 2     | 0    |
| Wind                                                                          | WSAvg       | 14.9   | 13.7  | 14.4  | 14.9  | 17.0  | 17.7  | 16.4  | 14.7  | 13.8  | 14.1  | 14.9  | 14.1  | 13.7 |
| Precipitation                                                                 | PrecAvg     | 34.3   | 2.1   | 2.1   | 2.6   | 2.1   | 3.3   | 2.6   | 3.1   | 2.0   | 5.9   | 4.2   | 2.8   | 1.7  |
|                                                                               | PrecMax     | 50.0   | 6.3   | 5.3   | 5.4   | 8.6   | 12.4  | 8.9   | 13.8  | 7.2   | 16.6  | 14.1  | 10.0  | 6.0  |
|                                                                               | PrecMin     | 18.2   | 0.0   | 0.1   | 0.1   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 1.3   | 0.0   | 0.0   | 0.3  |
|                                                                               | PrecStd     | 9.5    | 1.8   | 1.7   | 1.7   | 2.0   | 3.3   | 2.2   | 3.9   | 1.8   | 3.7   | 3.9   | 2.7   | 1.5  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 70.2  | 70.7  | 74.8  | 82.4  | 82.4  | 85.5  | 86.2  | 87.2  | 88.4  | 85.1  | 79.2  | 74.5 |
|                                                                               |             | MCWB   | 62.5  | 65.6  | 64.2  | 60.9  | 73.9  | 78.8  | 78.4  | 78.6  | 76.3  | 78.1  | 73.5  | 69.1 |
|                                                                               | 2%          | DB     | 68.3  | 68.8  | 72.7  | 77.7  | 81.4  | 84.5  | 85.4  | 86.3  | 86.2  | 84.0  | 77.9  | 72.5 |
|                                                                               |             | MCWB   | 65.1  | 66.2  | 68.4  | 73.4  | 77.1  | 78.2  | 78.5  | 78.7  | 78.0  | 77.8  | 72.9  | 69.5 |
|                                                                               | 5%          | DB     | 67.3  | 67.6  | 71.3  | 76.5  | 80.9  | 84.1  | 84.9  | 85.7  | 85.4  | 83.0  | 77.0  | 71.3 |
|                                                                               |             | MCWB   | 64.7  | 65.5  | 68.6  | 73.4  | 76.7  | 78.0  | 78.4  | 78.8  | 78.4  | 77.2  | 72.3  | 68.9 |
|                                                                               | 10%         | DB     | 66.1  | 66.4  | 70.1  | 75.4  | 80.3  | 83.6  | 84.4  | 85.2  | 84.9  | 82.0  | 75.8  | 70.1 |
|                                                                               |             | MCWB   | 64.0  | 64.4  | 67.9  | 72.8  | 76.3  | 77.7  | 78.2  | 78.8  | 78.1  | 76.3  | 71.4  | 67.9 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 68.0  | 68.6  | 71.9  | 77.1  | 79.9  | 82.1  | 81.6  | 82.4  | 81.6  | 80.5  | 76.3  | 72.0 |
|                                                                               |             | MCDB   | 68.4  | 69.1  | 72.6  | 77.3  | 80.6  | 83.8  | 84.0  | 84.7  | 84.9  | 83.0  | 78.6  | 73.1 |
|                                                                               | 2%          | WB     | 66.9  | 67.6  | 70.9  | 75.8  | 79.0  | 80.8  | 80.6  | 81.8  | 80.5  | 79.4  | 74.7  | 70.9 |
|                                                                               |             | MCDB   | 67.5  | 68.3  | 71.6  | 76.5  | 80.2  | 83.3  | 83.6  | 84.4  | 84.4  | 82.6  | 76.8  | 72.2 |
|                                                                               | 5%          | WB     | 66.0  | 66.6  | 69.8  | 74.7  | 78.3  | 79.9  | 80.2  | 81.0  | 80.0  | 78.3  | 73.5  | 69.8 |
|                                                                               |             | MCDB   | 66.7  | 67.3  | 70.6  | 75.7  | 79.9  | 83.0  | 83.5  | 84.2  | 84.3  | 81.9  | 76.0  | 71.0 |
|                                                                               | 10%         | WB     | 65.0  | 65.5  | 68.9  | 73.4  | 77.6  | 79.1  | 79.6  | 80.3  | 79.3  | 77.5  | 72.5  | 68.7 |
|                                                                               |             | MCDB   | 65.9  | 66.2  | 69.8  | 74.6  | 79.5  | 82.7  | 83.3  | 84.0  | 84.0  | 81.4  | 75.3  | 70.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 8.6   | 7.5   | 6.0   | 4.4   | 3.2   | 2.8   | 2.8   | 3.0   | 5.2   | 6.3   | 8.0   | 9.2  |
|                                                                               |             | MCDBR  | 8.5   | 7.4   | 6.6   | 5.9   | 3.8   | 3.3   | 3.0   | 3.1   | 4.7   | 4.8   | 7.0   | 7.8  |
|                                                                               |             | MCWBR  | 8.7   | 6.7   | 6.0   | 5.0   | 3.6   | 3.0   | 2.8   | 2.8   | 3.7   | 4.7   | 7.5   | 8.4  |
|                                                                               | 5% WB       | MCDBR  | 7.2   | 5.8   | 3.4   | 3.4   | 3.0   | 3.2   | 2.6   | 2.6   | 4.2   | 5.5   | 6.0   | 6.9  |
|                                                                               |             | MCWBR  | 8.6   | 6.6   | 4.1   | 3.9   | 3.6   | 3.4   | 2.8   | 2.8   | 4.0   | 6.2   | 7.1   | 8.2  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.364  | 0.371 | 0.388 | 0.434 | 0.475 | 0.451 | 0.469 | 0.453 | 0.441 | 0.394 | 0.380 | 0.353 |      |
|                                                                               | taud        | 2.484  | 2.467 | 2.414 | 2.290 | 2.207 | 2.341 | 2.306 | 2.351 | 2.345 | 2.473 | 2.464 | 2.530 |      |
|                                                                               | Ebn at Noon | 279    | 286   | 287   | 275   | 262   | 267   | 262   | 266   | 267   | 274   | 271   | 277   |      |
|                                                                               | Edh at Noon | 30     | 33    | 36    | 43    | 46    | 40    | 42    | 40    | 39    | 32    | 30    | 28    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 947    | 1125  | 1449  | 1728  | 1947  | 2135  | 2097  | 2006  | 1658  | 1446  | 1082  | 852   |      |
|                                                                               | RadStd      | 105    | 184   | 152   | 164   | 131   | 110   | 173   | 108   | 140   | 144   | 114   | 113   |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +146  |

Nomenclature: See separate page